Predictive Analytics and Organizational Architecture: Plant-Level Evidence from Census Data

by

Eva Labro
University of North Carolina at Chapel Hill, Kenan-Flagler Business School

Mark Lang
University of North Carolina at Chapel Hill, Kenan-Flagler Business School

Jim Omartian
University of Michigan, Ross School of Business

CES 19-02 January, 2019

The research program of the Center for Economic Studies (CES) produces a wide range of economic analyses to improve the statistical programs of the U.S. Census Bureau. Many of these analyses take the form of CES research papers. The papers have not undergone the review accorded Census Bureau publications and no endorsement should be inferred. Any opinions and conclusions expressed herein are those of the author(s) and do not necessarily represent the views of the U.S. Census Bureau. All results have been reviewed to ensure that no confidential information is disclosed. Republication in whole or part must be cleared with the authors.

To obtain information about the series, see www.census.gov/ces or contact Christopher Goetz, Editor, Discussion Papers, U.S. Census Bureau, Center for Economic Studies 5K028B, 4600 Silver Hill Road, Washington, DC 20233, CES.Working.Papers@census.gov. To subscribe to the series, please click here.

Abstract

We examine trends in the use of predictive analytics for a sample of more than 25,000 manufacturing plants using proprietary data from the US Census Bureau. Comparing 2010 and 2015, we find that use of predictive analytics has increased markedly, with the greatest use in younger plants, professionally-managed firms, more educated workforces, and stable industries. Decisions on data to be gathered originate from headquarters and are associated with less delegation of decision-making and more widespread awareness of quantitative targets among plant employees. Performance targets become more accurate, long-term oriented, and linked to company-wide performance, and management incentives strengthen, both in terms of monetary bonuses and career outcomes. Plants increasing predictive analytics become more efficient, with lower inventory, increased volume of shipments, narrower product mix, reduced management payroll and increased use of flexible and temporary employees. Results are robust to a specification based on increased government demand for data.

Keyword*:*

JEL Classification:

* This research was conducted while the authors were Special Sworn Status researchers of the U.S. Census Bureau at the Triangle Federal Statistical Research Data Center (FSRDC) and the Michigan FSRDC. Any opinions and conclusions expressed herein are those of the authors and do not represent the views of the U.S. Census Bureau. All results have been reviewed to ensure that no confidential information is disclosed. We thank Raffi Indjejikian, Venky Nagar, Stephen Glaeser and workshop participants at the University of Michigan, the University of Colorado and the University of North Carolina at Chapel Hill for helpful comments.

1. Introduction

One of the most significant trends in manufacturing in recent decades has been technology-driven enhancements in the ability to gather, store and process large amounts of data quickly, efficiently and at reasonable cost.[1] With rapid increases in the quantity of available data, companies are increasingly turning to predictive analytics—"statistical algorithms and machine learning techniques to identify the likelihood of future outcomes based on historical data"[2]—to glean potential operational improvements. For example, plants are able to gather and process granular real-time data to optimize manufacturing cycle time, equipment and labor effectiveness, supply chain effectiveness, overall plant efficiency, product quality and defect rates.

Discussions of the merits of predictive analytics typically focus narrowly on their potential to increase the ability to capture, transmit and analyze large datasets to improve process efficiency at the plant level (Dilda, Mori, Noterdaeme, and Schmitz, 2017; EY, 2014). Less discussed, but equally important, is the possibility that the information generated through predictive analytics and related big data techniques has the potential to fundamentally alter the optimal organizational design and control structure within the firm. Theories of organizational architecture hinge critically on the role of information asymmetry and information quality, which agency theory posits as shaping optimal delegation of decision-making between headquarters and the plant, performance evaluation and incentive design in the firm. As plants implement predictive analytics, they generate large amounts of data that can potentially alter the amount and nature of decision-

[1] For example, Deloitte (2018) notes, "The manufacturing industry as we know it is fundamentally changing, with advanced technologies increasingly underpinning global competitiveness and economic prosperity. Many leading 21st-century manufacturers are converging digital and physical worlds in which sophisticated hardware combined with innovative software, sensors, and massive amounts of data and analytics is expected to produce smarter products, more efficient processes, and more closely connected customers, suppliers, and manufacturers."

[2] www.sas.com/en_us/insights/analytics/predictive-analytics.html

facilitating information (used to support decision-making) or decision-influencing information (used to support performance measurement and incentives), or both (Demski and Feltham, 1976). This raises important questions such as: who in the organization decides what data are gathered and how they are used; does the availability of predictive hard data change the power dynamics between the plant and headquarters; does the additional information alter compensation design, incentive intensity and target setting practices for plant managers; and to what extent is plant efficiency affected?

Despite the potential for predictive analytics to interact importantly with organizational architecture, there is no large-sample evidence about their use in practice and, in particular, on how organizations adapt their internal architecture in response to the increase in internal information accompanying predictive analytics. Academic analysis of predictive analytics has been largely limited to conceptual papers suggesting ways that big data in general, and predictive analytics in particular, might be used to improve firms' operations in specific fields of interest.[3] Most empirical evidence is limited to case-based studies investigating outcomes of predictive analytics in very limited contexts.[4] However, we know of no large-scale studies focusing on the role of predictive analytics, likely due to the lack of available data on their use for a large, representative, sample of establishments.[5]

[3] For example, Alles (2015) discusses the potential role of big data on auditing; Schneider, Dai, Janvrin, Ajayi, and Raschke (2015) and Vasarhelyi, Kogan, and Tuttle (2015) on accounting; Baesens, Bapna, Marsden, Vanthienen, and Zhao (2016) and Agarwal and Dhar (2018) on information systems; Corbett (2018) on sustainability; and Guha and Kumar (2018), Sanders and Ganeshan (2018) and Singhal, Feng, Ganeshan, Sanders, and Shanthikumar (2018) on operations and supply management.

[4] For example, Akturk, Ketzenberg, and Heim (2018) provides evidence of improved management of sales and customer returns resulting from data analytics for a single retailer and Cui, Gallino, Moreno, and Zhang (2018) suggests that sales prediction at a retailer improved following implementation of predictive analytics. Evidence in the practitioner literature suggests that many companies experience limited returns to their investment in big data (Ross, Beath, and Quaadgras, 2013).

[5] Research such as Brynjolfsson, Hitt, and Kim (2011) and Brynjolfsson and McElheran (2016) investigates the effects of data-driven decision making more generally. We include controls for other forms of data gathering in our empirical specifications to better isolate the effects of predictive data analytics.

In this paper, we use new confidential data from the US Census Bureau to examine the increasing use of predictive data analytics, its relation to organizational architecture in the firm and associated plant-level outcomes. Census data have several unique advantages for our purposes. Most importantly, for the first time in 2015, the Management Organization and Practices Survey (MOPS) from the Census Bureau queried a large, representative sample of plants on their use of predictive analytics. Plants provided data, for 2010 and 2015, on how reliant they were on predictive analytics, as well as who (local managers or headquarters) chose what data was collected, and how frequently key data sources (e.g. performance indicators from production technology or feedback from managers) were used in the decision making process. Beyond our primary construct of interest, predictive analytics, the MOPS includes numerous questions on management practices, organizational hierarchy, decision-making, and uncertainty, which permit us to examine interactions among management control mechanisms. We additionally use the Annual Survey of Manufactures (ASM) and the Census of Manufactures (CMF), which permit us to measure important control variables, such as the value of product shipments, number and types of employees, number of products produced and plant age, as well as to investigate potential efficiency outcomes.

Our main sample includes more than 25,000 plants. Response rates are high because compliance with Census information requests is mandatory and personnel follow up repeatedly. Because responses are confidential (even from other branches of government) and there are penalties, there should be few reasons to misreport. The sampling is randomized and stratified to ensure that it covers a representative spectrum of plants not limited by industry, geography, size, or corporate type, and includes both publicly and privately held firms. In terms of identification, because we have observations for both 2010 and 2015, we can conduct our analysis based on

changes in the use of predictive analytics over time. Since the unit of analysis is the plant and there is a large number of plants in the sample, we can conduct analyses using firm- and plant-level controls, along with an extensive fixed effects structure. In our primary analyses we include firm fixed effects (along with plant industry and county fixed effects), so that comparison is across plants within a given firm, controlling for location and industry effects, which allows us to rule out a wide range of potential omitted variables such as firm-level changes, local economics and industry shocks.

We use the Census data to examine three primary sets of research questions:

1. To what extent is the rise in predictive analytics pervasive across plants and, closely related, in what contexts is it most pronounced? Our results indicate that nearly a third of sample firms report increased use in at least one of their plants between 2010 and 2015. Consistent with predictions, the increase is particularly pronounced in young plants, large firms with professional management, more educated work forces, and more stable industries.
2. How do predictive analytics interact with organizational architecture in the firm? Prior theoretical literature (Aghion and Tirole, 1997; Prendergast, 2002) suggests that informational asymmetries between the plant and headquarters play an important role in determining optimal delegation, performance measurement and incentive intensity.[6] To the extent that predictive analytics provide important, objective, information that was not previously available, we expect an effect on the balance of power between the plant and headquarters. To examine this effect, we decompose organizational

[6] For overviews of the literature on delegation and incentives, target setting, and the use of firm- versus plant-level performance measures more generally, see Hofmann and Indjejikian (2018, forthcoming), Bouwens and Kroos (2017), and Bouwens, Hofmann, and Van Lent (2018), respectively.

architecture into three components following Brickley, Smith, and Zimmerman (1995) and Ittner and Larcker (2001):

a. *Decision-making authority*. Are predictive analytics related to the delegation of decision rights for capital expenditures, marketing and human resource management to the plant? It is ex ante unclear whether predictive data analytics should facilitate or mitigate delegation. In particular, plant-level predictive analytics might increase the information advantage of plant managers and increase delegation or might enhance the ability of the headquarters to centralize decision-making and decrease delegation. Consistent with the latter possibility, our results indicate predictive analytics are associated with reduced delegation of decision making to plants. Further, headquarters tends to take the major role in specifying the specific data to be gathered, suggesting that headquarters uses data analytics to centralize decision-making. Overall, this indicates that predictive analytics provides decision-facilitating information.

b. *Performance evaluation mechanisms*. Is adoption of predictive analytics associated with changes in how performance objectives are defined and communicated? To the extent that the increased amount of information produced by predictive analytics increases overall information quality and reduces noise, we would expect increased use of quantitative data in incentive design. We find consistent evidence that predictive analytics are associated with increased awareness of quantitative targets throughout the plant. In addition, performance targets become more accurate and long-term oriented, reflecting increased availability of hard, forward-looking information.

c. *Incentive design and intensity*. Are predictive analytics associated with changes in the strength of explicit incentives? High-powered incentives reflect a trade-off between achieving desired outcomes and imposing risk on managers. Predictive analytics potentially increase information quality and reduce noise in performance evaluation, reducing the risk of incentives tied to performance measures. Consistent with that view, we find that increased use of predictive analytics is associated with higher-powered incentives for plant managers as reflected in stronger links between performance and bonuses as well as probability of promotion and speed of termination for underperformance. Overall, this suggests that predictive analytics also offer decision-influencing information.

3. Given that delegation, performance measurement and incentives change, is use of predictive analytics associated with operational changes at the plant? While we cannot measure overall plant profitability, our results suggest that predictive analytics are associated with increases in the efficiency of plant operations. Plants increase volume of shipments for a given level of capacity, while streamlining the range of products offered and decreasing excess inventory. Management payroll decreases, consistent with increased centralization reducing the need for highly-skilled local managers. Plants shift to more flexible staffing, with more temporary employees, nontraditional schedules and workers cross-trained to perform multiple tasks.

While results are robust to a wide range of controls, as well as firm, plant industry and county fixed effects, given the nature of the data we are cautious about making causal claims. We view the substantial increase in use of predictive analytics as likely reflecting the economy-

wide increase in the ability of firms to capture, transmit and analyze large plant-level datasets in a cost-efficient manner. However, it is possible that, for example, delegation and use of predictive analytics may have evolved in a codetermined process.[7] To provide further insight into the role of data-gathering, we employ an approach that exploits increases in data acquisition in response to government requirements, which are likely to be more exogenous to the firm. We find similar results in terms of reduced delegation, more accurate targets and greater sensitivity of firing to underperformance, suggesting it is more likely to be the availability of data that drives the management control mechanism design and incentive decisions.

We believe we make three primary contributions to the existing literature. First, and most directly, we provide initial evidence suggesting a link between predictive data analytics and organizational architecture. As noted earlier, the practitioner literature focuses on the benefits of data analytics for plant-level efficiency, but our results suggest that increased prevalence of plant-level predictive analytics has important implications for the power structure and incentives in the firm more generally. Predictive analytics alter the nature of the internal information environment in that they provide both improved decision-facilitating as well as decision-influencing information.

Second, we provide empirical evidence more generally evaluating predictions from theoretical research for understanding the role of information acquisition in decision rights assignment, performance evaluation and incentive design. Our data provide a setting in which the

[7] It seems unlikely that causality is reversed (e.g., shocks to optimal delegation drive changes in data analytics). Other than availability of information (which is our focus) delegation is likely a function of uncertainty at the headquarters or plant. We find no evidence to suggest significant changes in uncertainty for the firms or plants during our sample period, and we include controls for likely drivers of plant- and headquarters-level uncertainty in our regressions, including plant industry and county fixed effects.

predictions from that literature can be examined based on a significant change in available information for a large and representative sample of headquarter and plant parings.[8]

Finally, and more indirectly, our results provide insight into potential opportunities for managers more generally as data and analysis become increasingly pervasive, especially in multi-locational firms. Taken at face value, our results suggest that increases in available data will likely facilitate reductions in delegation and increases in reliance on targets and other quantitative data in incentives. Our results on outcomes suggest the opportunity for significant efficiency increases with additional data analytics, but also shifts in the workforce toward more temporary and on-demand employment.

There are, of course, important caveats. First, as with all studies in this literature, we cannot observe the counterfactual (i.e., features of the plant had it not implemented predictive analytics). Relative to the prior literature, an advantage of the Census data is that we can conduct our analysis in changes, while including fixed effects for the firm, county and industry sector, as well as numerous controls for economic factors such as plant size, complexity, education levels and plant hierarchy to rule out alternative explanations

Second, the 2010 Census variables are self-reported "recall" data from 2015. Given "anchoring bias" (a tendency to rely too heavily on more recent salient data in recalling past events (Furnham and Boo, 2011)), our results may understate the increased use of predictive analytics.[9]

[8] While, to our knowledge, there is no large-scale empirical research examining predictive analytics, Bresnahan, Brynjolfsson, and Hitt (2002) finds that information technology (IT) adoption more generally is associated with greater decentralization, opposite to our results. Bloom, Garicano, Sadun, and Van Reenen (2014) contrast Enterprise Resource Planning adoptions with data intranet adoptions and argue that the nature of the IT determines whether centralization or decentralization obtains. The IT literature does not study the relation between IT adoption and performance measurement or incentive design.

[9] There are also advantages associated with recall data. The fact that data for both 2015 and 2010 are being reported on the same form, and likely by the same employee at the plant, increases the probability that definitions for constructs such as "predictive data analytics" are being applied consistently over time.

However, our primary interest is in cross-sectional differences in the use of predictive analytics, so any systematic bias should not affect conclusions (although noise may reduce statistical power).

Third, as with many studies, the link between the theoretical variables and empirical constructs is imperfect and we use pre-structured archival data so we do not control the questions included in the survey. That said, the MOPS data questions were specifically designed by a team of economists from the Census Bureau collaborating with academic economists to capture the underlying theoretical constructs and have been validated in other studies.

2. Background and Hypotheses

In many ways, the setting of the typical manufacturing firm falls neatly into the structure of agency theory. In particular, the headquarters and plant managers have different incentives, and inherent information asymmetry exists because the plant is typically not collocated with the headquarters and plant managers have an informational advantage with respect to local operations in decision-making. The degree to which headquarters can measure performance and incentivize effort depends on the quality and amount of available information. Given potential issues with information asymmetry and information quality, headquarters faces trade-offs in deciding on an optimal level of delegation, performance monitoring and incentive design. Predictive analytics potentially change the calculus by providing a large amount of tangible, forward-looking data. Although our analysis is somewhat descriptive and inherently limited by the available data, we structure our discussion around five primary hypotheses based on predictions, along with plausible tension, from the organizational architecture literature.

Determinants of Predictive Analytics Use

In our first set of analyses, we consider the features of firms that significantly expanded their use of predictive analytics between 2010 and 2015. We view this analysis as descriptive but important to understanding the drivers of predictive analytics for our later analyses. We expect that adoption of predictive analytics will be concentrated in plants for which likely benefits are higher, adoption costs are lower and more financial resources are available. In terms of benefits, because data analytic techniques use historical data to predict future outcomes, they are likely to be more effective when the outputs to the firm are subject to less uncertainty. In terms of costs, predictive analytics are likely to rely on implementation by sophisticated employees, so we expect greater use by plants with a more highly educated workforce (Brynjolfsson and McElheran, 2016; Tambe, 2014). In terms of available resources, we expect greater use of predictive analytics by plants of firms that are larger and not family-owned since those firms are more likely to have access to the resources to effectively implement predictive analytics.

H1: Adoption of predictive analytics is greater for plants with more stable operations, more access to resources and a more educated work force.

Interplay between Predictive Analytics and Organizational Architecture

The second set of three hypotheses is more substantive and deals with the interaction between predictive analytics and the firms' organizational architecture. To structure our predictions, we rely on the three-pronged framework for organizational architecture discussed in (Brickley et al., 1995) and Ittner and Larcker (2001): (1) delegation of decision rights, (2) performance monitoring, and (3) incentive strength.

First, in terms of *delegation of decision rights*, we expect the availability of predictive analytics to affect the degree to which the firm centralizes decision making between headquarters and local plant managers. Prior theoretical literature suggests that delegation of authority will be greatest in cases in which there is greater information asymmetry between headquarters and plants, offering the plant the opportunity to react more nimbly to changes in their local environment given their superior information access.[10] However, delegating decision-making authority to plants increases the agency problem between headquarters and local management, as plant managers may make decisions that are in their best interest but not in the best interest of the firm.[11] Predictive analytics create quantitative "hard" data that have the potential to alter the information balance between plants and headquarters. The direction in which the information balance is altered depends, at least in part, on who (headquarters or plant managers) determines the specific data to be collected. However, to the extent predictive analytics' data are available to headquarters, it likely reduces the local manager's informational advantage, so we expect adoption of predictive analytics to be associated with less delegation of decision-making to plants.

H2: Predictive analytics are associated with a reduction in delegation of decision-making rights to the plant.

In addition, the availability of predictive analytics data potentially affects *performance monitoring* practices for plant managers, as information quality increases. In particular, the availability of hard data from the plant creates the capacity to increasingly tie plant-level incentives to explicit quantitative targets. As a result, we expect predictive analytics to be associated with increased awareness of explicit quantitative targets at the plant. Further, because the forward-looking character of predictive analytics provides better ability to forecast, we expect incentives

[10] See, for example, Abernethy, Bouwens, and Van Lent (2004) and Bushman, Indjejikian, and Penno (2000).
[11] See, for example, Nagar (2002) and Prendergast (2002).

to focus on longer-term targets. Finally, because plants are better able to predict future outcomes, we expect targets to be more "accurate" in the sense that they reflect reasonably attainable goals. Prior research suggests that setting targets too low or too high relative to what is reasonably attainable reduces their incentive effect (Eyring and Narayanan, 2018; Webb, Williamson, and Zhang, 2013). We expect that predictive analytics facilitate setting targets that are neither too easy nor unrealistically difficult, reducing noise and bias in performance measurement.

H3: Predictive analytics are associated with increased awareness of targets, as well as targets that are more forward-looking and accurate.

Third, we expect predictive analytics to affect *incentives* in terms of both nature and strength. First, we consider the strength of the incentive effects more generally. If predictive analytics increases information quality, performance measure quality will improve. As a result, the strength of incentives should increase because objective information is more readily available on which to contract, reducing the risk premium in incentive compensation (Holmström, 1979). As a result, we expect the availability of predictive analytics to increase the sensitivity of compensation and career-outcomes to performance outcomes.[12]

Second, we consider the association between predictive analytics and the choice of whether to tie incentives to plant- or firm-level performance. To the extent that availability of hard predictive data increases centralization of decision-making in headquarters, the firm will attempt to tie plant managers' incentives more closely to firm-level outcomes, reflecting headquarters'

[12] Potentially offsetting, if predictive analytics increase centralization, this may reduce the need to rely on strong incentives to control the agency problem because of the joint determination of delegation and incentive compensation (Moers, 2006; Nagar, 2002; Prendergast, 2002).

desire to coordinate plant managers' actions (Bushman, Indjejikian, and Smith, 1995). As a result, we expect a positive association between plant manager incentives and firm-level performance.[13]

H4: Predictive analytics are associated with higher-powered bonus, promotion and dismissal incentives, and with incentives that are tied more strongly to firm-level performance.

Predictive Analytics and Outcomes

Our final analyses are based on plant-level outcomes. Because the Census does not provide income statements for plants, it is difficult to assess overall outcomes such as plant-level profitability. However, we can speak to narrower outcomes related to the internal organization of the workforce and the production processes.[14] In particular, given that predictive analytics shift decision-making to headquarters, we expect a reduction in spending on local plant managers. In addition, we expect improved optimization of production processes in plants that are able to increase production for a given level of plant capacity, streamline their range of products and reduce inventory levels. Finally, we expect predictive analytics to facilitate flexible manufacturing, resulting in greater reliance on temporary workers, flexible scheduling and multifunctional employees.

H5: Predictive analytics are associated with more efficient management of production and inventory, and more flexible staffing.

[13] Providing tension, predictive analytics may make *plant-level* performance measures more informative and therefore more useful for contracting (Holmström, 1979). The empirical evidence on the level at which incentives are offered (firm versus plant) is mixed (Bouwens et al., 2018; Bouwens and Van Lent, 2007; Bushman et al., 1995).

[14] In particular, implementing data systems likely requires significant resource expenditures that we do not observe. In addition, we only observe outcomes for the plant, so centralizing decision-making may require additional resources at headquarters or might impose other unobservable costs (or benefits) to the headquarters or plant.

3. Data

Our primary data come from three confidential datasets gathered by the US Census Bureau. While known primarily for its decennial tabulation of individuals, the Census also gathers highly detailed data on US economic activity. Census publishes aggregates of the economic data for the benefit of policy makers and businesses making operational decisions, but researchers on approved projects may apply for access to the underlying microdata. We focus on manufacturing activity in this study because of its critical role in the US economy and the fact that, as a result, the Census Bureau collects very rich, highly granular data on the sector. We use the CMF and ASM data to measure plant-level characteristics and the MOPS data to measure aspects of the managerial control systems employed at the plant.

CMF and ASM Data

On an annual basis, the Census Bureau gathers detailed information about domestic manufacturing establishments—both privately and publicly held.[15] In years ending in "2" or "7", the Census Bureau conducts a full Census of Manufactures (CMF) and sends surveys to roughly 168,000 establishments, representing all but the smallest manufacturing plants (single-plant companies with fewer than 20 employees).[16] Plants are required to provide operational data, including sales, employees, payroll, inventories, capital expenditures, operating expenses, and additional background information. The forms do not ask directly about accounting earnings.[17]

[15] These data underpin a number of economic indicators such as the Federal Reserve's Industrial Production index, the Bureau of Labor Statistics' producer price index, and the Bureau of Economic Analysis's input-output tables.
[16] The Census defines an establishment as "a single physical location where business is conducted or where services or industrial operations are performed." For manufacturers, an establishment corresponds to an individual production plant. For detailed information on the CMF and ASM process, refer to https://federalregister.gov/a/2011-5990.
[17] Because the Census Bureau's aim is to derive aggregate manufacturing activity, many expenses such as Selling, General and Administrative Expense are not required to be reported at the plant level.

Outside of the quinquennial censuses, the Census Bureau surveys a subset of manufacturing establishments on an annual basis—the Annual Survey of Manufactures (ASM). Two years after each economic census, the Census Bureau selects a stratified sample of roughly 51,000 establishments: 33,000 plants from multi-establishment firms and 18,000 from larger single-establishment firms in the CMF. Once selected, this sample receives the ASM form each year for five years, after which a new sample is selected based on the subsequent CMF. The ASM questions are virtually identical to the main CMF questions.

The ASM's stratified sample is designed to capture the most significant economic activity and have adequate coverage of each industry and geographic area. As a result, roughly 15,400 of the largest plants are selected with certainty, accounting for roughly 67% of the economic activity in the CMF. The remaining sample of about 35,600 plants is selected using stratified sampling probabilities for each combination of geography, industry, and employment with the goal of providing representative data for the population of US manufacturing plants. A large portion of plants persist from one five-year sample to the next and the sample of firms surveyed annually remains constant during each five-year window. For both the CMF and ASM, plant responses are required by law. Additionally, the Census aggressively follows up with establishments that do not respond to minimize selection bias, resulting in a response rate of 70-80%.[18] Because responses are confidential, immune from legal process, and can only be used by the government for statistical purposes, there should be little incentive to misreport.[19]

[18] Non-response is likely a function of the fact that the establishments surveyed include some for which "eligibility could not be determined" (i.e., plants did not respond because they were not required to respond). In addition, there is likely attrition due to factors such as relocations or recent plant closing.

[19] Researchers granted access to the underlying Census microdata are not permitted to use it to make policy recommendations, eliminating another potential incentive for respondents to misreport.

MOPS Data

In 2015, the Census Bureau sent a supplemental survey alongside the ASM. The Management and Organizational Practices Survey (MOPS) includes 46 questions: 16 on management practices, 7 on organizational hierarchy, 6 on data and decision making, 8 on uncertainty and 9 on background information.[20] We rely on MOPS responses to measure the use of predictive analytics at the plant level. The MOPS also provides us with information about delegated decision rights, targets, and incentives. For each question, the form directs respondents to provide answers for 2010 and 2015. Most questions ask respondents to select from a list of choices. Thus, many of our variables are either binary or converted to a uniform 0-1 scale. About 52,000 plants were surveyed and 71% responded. While this survey provides unparalleled detail for such a large and broad sample of plants, data for predictive analytics are only available for 2010 and 2015. Because the ASM and MOPS were sent to the same plants, we have very high overlap between the two surveys and require responses from both for inclusion in our sample. This restriction leaves us with roughly 26,500 individual plants in our main sample.

Given the extensive disclosure requirements and procedures enforced by the Census, we are limited as to the extent of descriptive statistics we can report for our sample. Table 1 presents select descriptive statistics for our sample plants over the 2008-2016 window. The average plant shipped just under $35 million in product per year ($e^{10.46} \times 1000$), and produced 3.25 main products ($e^{1.18}$). Average management compensation was 7.1% of product shipments and staff compensation was 10.5%. The average hourly wage for production workers was $21.17.

[20] As with the ASM, plants were required by law to complete the MOPS, which was estimated to take about 45 minutes. Details can be found at: https://www2.census.gov/ programs-surveys/mops/technical-documentation/questionnaires/ma-10002_15_final_3-2-16.pdf?#

Insert Table 1 Here

Key Constructs

We draw our primary constructs of interest from Section C of the MOPS, "Data and Decision Making." Our measure of predictive analytics derives from responses to the question: "How frequently does this establishment typically rely on predictive analytics (statistical models that provide forecasts in areas such as demand, production, or human resources)?" Respondents select one frequency—never, yearly, monthly, weekly, or daily—individually for 2010 and 2015. We construct two measures of reliance on predictive analytics from these responses. *UsePA* is an indicator equaling one if the respondent chose an interval other than never. *PAScore* converts these responses to a 0-1 scale (0 = never, 0.25 = yearly, 0.5 = monthly, 0.75 = weekly, and 1 = daily) to capture the intensity with which plants use predictive analytics.

In terms of measuring delegated decision rights, we use a number of questions from the MOPS. Plants report the extent to which local management can make operational decisions without headquarters review on six dimensions: hiring, deciding large pay raises, new product introductions, advertising, pricing, and capital expenditures. Respondents answer the first five by indicating either that local management makes the decision (encoded as 1), headquarters makes the decision (encoded as 0) or headquarters and local management jointly make the decision (encoded as 0.5). For the capital expenditure question, plants report requisitioning authority thresholds for local managers; we encode under $1,000 as 0, $1,000-$9,999 as 0.25, $10,000-$99,999 as 0.5, $100,000-$999,999 as 0.75, and over $1 million as 1. We sum the scores from the six questions, creating an operational decision-making delegation measure, *DelegationScore*, ranging from 0-6.

A second category of delegation is the extent to which local management has control over the data and metrics collected. To capture this type of authority, we use two indicator variables *LocalChooseData* and *HQChooseData*, which encode responses to the MOPS question that asks which parties chose the information collected at the plant. *LocalChooseData* is 1 if the respondent checked that "Managers at this establishment" chose the type of data to be collected, and *HQChooseData* is 1 if the respondent checked "Managers at headquarters and/or other establishments" chose the type of data to be collected.[21]

We measure a variety of aspects of targets used for performance evaluation. We first capture the use of various target horizons. *ShortTermTargets* is an indicator if the respondent checked "Main focus was on short-term (less than one year) production targets," whereas *LongTermTargets* is an indicator if the respondent checked either "Main focus was on long term (more than one year) production targets," or "combination of short-term and long-term production targets." We next turn to the awareness of targets by plant employees, *TargetAwareness*, which is scaled from 0-1, where 0 equates to "only senior managers" being aware of targets, 0.33 is "most managers and some production workers," 0.67 is "most managers and most production workers," and 1 is "all managers and most production workers." We also capture the difficulty of targets used, both based on the respondent's subjective assessment of the effort required to achieve the target and on whether or not the targets were actually achieved. We use five indicator variables: *LittleEffort, SomeEffort, NormalEffort, MoreThanNormalEffort,* and *ExtremeEffort*, based on how much effort the respondent said was necessary to achieve the target. By using separate indicators as opposed to a linear difficulty scale, our models are able to accommodate nonlinearities in target difficulty. In particular, we expect targets to be less effective if they require either too much or

[21] Because respondents check all that apply, *LocalChooseData* and *HQChooseData* may both be 1 for a given plant.

too little effort (Eyring and Narayanan, 2018). *MeetOrBeat* is an indicator that takes on a value of one if the plant achieved the target.

We measure incentives using the existence of monetary bonuses and career trajectory outcomes. *UseBonus* is an indicator if plant management has a performance-based bonus plan. *Terminations* is the speed at which underperforming managers are terminated or reassigned (set to 1 if less than 6 months, 0.5 if greater than 6 months, and 0 if rarely or never). *Promotions* captures how strongly promotion criteria are based on performance or ability (set to 1 if promotions are based solely on performance or ability, 0.67 if based partly on performance or ability and partly on other factors like tenure or family connections, 0.33 if based mostly on other factors, and 0 if managers are typically not promoted).[22] Turning to incentive criteria, *CompanyBonus* is an indicator if managers at the plant have bonuses available based on company-wide performance; *PlantBonus* is an indicator if they are based on plant-specific performance.[23]

Research Design

We estimate predictive analytics use as a function of firm and plant characteristics, and then estimate organizational architecture and outcomes as a function of predictive analytics usage. For each analysis, we employ models based on 2015 levels, and the changes between 2010 and 2015. Because our primary interest is the association between the implementation of predictive analytics and other plant-level variables, our focus throughout is the changes specification. However, we estimate models in levels using 2015 data as well and tabulate coefficients for select analyses. Unless indicated, our inferences are the same for both the levels and changes analyses.

[22] Campbell (2008) points out that promotions are an underresearched instrument of providing incentives.
[23] Because respondents check all that apply, *CompanyBonus* and *PlantBonus* are not mutually exclusive.

In addition to employing a changes specification, we report our analyses both with and without a battery of fixed effects. Firm fixed effects are powerful in this context because they permit us to focus on within-firm plant level variation in predictive analytics adoption abstracting from firm-level effects.[24] Coupled with the changes specification, we are able to isolate the relation between the increased use of predictive analytics and, for example, delegation or incentive design for a specific plant controlling for overall firm-level changes in predictive analytics and firm-level changes in delegation and incentive design, eliminating a wide range of potential omitted correlated variables. The large number of observations also allows us to include fixed effects for the county and industry of the specific plant. Use of industry fixed effects (primary 4-digit NAICS industry of the plant) mitigates concerns about general changes in sector-level economic conditions during our sample period.[25] County fixed effects mitigate potential concerns that changes in economic conditions in specific locations (e.g., economic growth or unemployment rates) could affect management control structures.[26] Because the sample includes a significant number of single-plant firms, we also report results without firm fixed effects to provide evidence for the population more generally. In all of our analyses, we cluster standard errors at the firm level.

In our setting, we argue that the primary causal direction is most likely that predictive analytics usage, driven by reductions in implementation costs and improvements in effectiveness, affects key organizational architecture constructs. However, in practice organizational

[24] To our knowledge, Campbell, Datar, and Sandino (2009) is the only other study in the accounting literature on delegation that has been able to focus on within-firm variation (although they are not able to observe changes). They use data from convenience store chains to understand why some stores are franchised while others are owned outright.
[25] The plant can be in a different industry than the primary industry of the parent firm.
[26] One potential concern is that the plants adopting predictive analytics also changed in other ways (e.g., robotics or other fundamental changes to the production process). As discussed in more detail in Section 5, there is no evidence that overall staff pay changed, inconsistent with an automation explanation. In addition, regressions include controls for payroll growth, as well as for management and staff education, and are robust to controls for capital expenditures.

architecture and predictive analytics usage may arise from a co-determined process. To mitigate this possibility, we also employ an analysis in which we instrument for the increased reliance on data analytics using increases in government-mandated information collection. The basic logic is as follows. For plants to follow government-imposed mandates, investments must be made in systems (technological or organization-based) that gather and process the required information. These systematic improvements to fulfill the government's requirements also lower the incremental cost of collecting and processing other operational data, making the use of predictive analytics more attractive. However, the government-mandated data collection is less likely to have a direct effect on the organizational architecture of the firm. To capture government mandated data collection, we use changes in an indicator of whether or not the government chose the type of data collected at the plant.

4. Results and Discussion

All results using confidential Census data undergo a review by Census disclosure personnel to insure there is no risk of inferring individual responses. Because each specific number reported is vetted by Census personnel to ensure confidentiality, we attempt to minimize the nonessential quantitative data tabulated, limiting reporting of, for example, coefficients on control variables to early tables and noting that they are qualitatively similar in later analyses.

Insert Table 2 here.

Table 2 presents descriptive statistics for key variables of interest. In 2015, 72.4% of manufacturing plants were using predictive analytics (*UsePA* = 0.724) and in terms of intensity, the average plant relied on predictive analytics on slightly longer than a monthly interval (*PAScore* = 0.478). These levels represent a substantial increase relative to 2010. Focusing on *PAScore*, which captures both changes in the extensive and intensive margins, we see an increase of 0.087

from 0.391, which equates to the average plant roughly moving from a quarterly to a monthly usage frequency. Tallying all plants that logged an increase, 23.1% report heavier reliance on predictive analytics. Counting firms where at least one plant reported an increase in its *PAScore*, 30.1% reporting an increase for at least one plant.

Determinants of Data Acquisition

With respect to our first research question, we provide evidence on determinants of the use of predictive analytics in Table 3. While this analysis is descriptive, there is not, to our knowledge, other broad sample evidence on use of predictive analytics. We report results for both levels and changes but, in subsequent analyses, focus on changes to better assist with identification.

In Table 3, Panel (a), we report 2015 levels analyses with both *UsePA* and *PAScore* as dependent variables. Results are consistent across both measures. In terms of firm-level determinants, plants are more likely to rely on predictive analytics if the parent firm is larger (i.e., larger payroll, multinational or multi-plant firm). In addition, family-run firms rely less on predictive analytics, consistent with lower managerial sophistication and fewer resources. Younger firms are also more likely to rely on predictive analytics, perhaps reflecting the fact that young firms tend to be more innovative because they do not have to navigate existing organizational routines and structures (Henderson and Clark, 1990).

Insert Table 3 here.

Because predictive analytics use historical data to forecast, they are less likely to be effective for industries that are innovating and changing rapidly. Consistent with that intuition, we observe a negative relation between use of predictive analytics and both industry growth and technological intensity. Similar to the firm-level evidence, newer and larger plants are more likely

to rely on predictive analytics, as are plants in which more managers and staff hold college degrees. Columns 3 and 4 report results with firm, industry, and county fixed effects, focusing on variation within firm and controlling for differences in industry and local economic conditions. Results are consistent in suggesting that, within firms, the plants that are likelier to employ predictive analytics are larger with better-educated staff (although manager education is insignificant).

Panel (b) reports results for changes in the *UsePA* and *PAScore* based on changes in plant characteristics, both with and without firm fixed effects. Results are generally consistent with expectations and with the results in Panel (a). Use of predictive analytics increased for growing firms and for firms with increased education levels for both managers and staff.[27]

Finally, results in Panel (c) provide evidence on complementarities between predictive analytics and other information sources. Predictive analytics are only one of several information sources available to managers and the 2015 MOPS also reports the extent to which firms rely on "performance indicators from production technology or instruments" and "formal or informal feedback from managers." Our interest here is in understanding the extent to which predictive analytics appears to be part of a more comprehensive data-driven strategy within the plant and, more importantly, whether predictive analytics are separable from other forms of data acquisition. We create two measures for each of these two data sources analogous to *UsePA* and *PAScore*: *UseProdPerfInd*, *UseMgrFdbck*, *ProdPerfIndScore*, and *MgrFdbckScore*. We then regress our predictive analytics measures on the corresponding measures from the other information sources, and the control variables from Panels (a) and (b), both with and without fixed effects, and in levels and changes. For parsimony, we only tabulate the coefficients for the information sources.

[27] As discussed later, the result for management and staff education may be, at least in part, endogenous in that plants planning to implement advanced data techniques hire more sophisticated employees to facilitate the process.

Results in Panel (c) indicate a positive relation between use of predictive analytics and use of both performance indicators from production technology and managerial feedback, consistent with the notion that plants that explicitly incorporate quantitative and qualitative input into decision-making are more likely to adopt predictive analytics. The fact that the link is stronger between predictive analytics and quantitative performance indicators also suggests that performance indicators may be gathered, at least in part, to facilitate estimation of predictive analytics models.[28] Importantly, however, the coefficient estimates on the other data sources are substantially less than one and the explanatory power of the regressions excluding fixed effects (but including controls) is 0.077 and 0.156 in Columns (5) and (7), suggesting that the relation between predictive analytics and other data sources is empirically separable. To ensure that our predictive analytics measure is not capturing other aspects of data gathering, we include controls for performance indicators and managerial feedback in all of our empirical analyses.

Data Acquisition and Delegation

Given the preceding evidence of a significant increase in predictive analytics usage for US manufacturing plants between 2010 and 2015, we now turn to understanding the interplay between data acquisition and organizational architecture. We begin by investigating the relation between predictive analytics and delegation of decision-making rights. Even if predictive analytics are intended primarily to increase efficiency at the plant, the availability of the resulting data has the potential to change the power balance between the plant and headquarters. The direction of the

[28] Despite the fact that we control for performance indicators from production technology or instruments in our analyses, in practice it may be difficult to separate out the availability of the data from their use in predictive analytics. Results are consistent if we combine our predictive analytics and performance indicator variables to measure overall "hard data".

effect depends in large part on whether the data generated serves as a substitute or complement to the private information held by the local manager.[29]

Table 4, Panel (a) reports results relating the change in predictive analytics to changes in delegation (*ΔDelegationScore*). Delegation is measured based on the plant's discretionary authority over decisions including human relations (hiring, promotion and pay), marketing (advertising, pricing and new product introduction) and capital expenditures. Because plants that are co-located with headquarters do not respond to the delegation questions, the sample size is smaller for this analysis and limited to remote plants. While our primary interest is in predictive analytics, we include the other data variables to ensure that we are not capturing the effects of data gathering or usage more generally.

Insert Table 4 here.

Results in Panel (a) strongly support the conclusion that predictive analytics use is associated with reductions in delegation of decision-making to the plant. The coefficient estimate for predictive analytics is negative and statistically significant, both with and without fixed effects. The effect of predictive analytics is substantially larger than for managerial feedback (which is never significant) and production performance indicators (which are only significant without fixed effects), suggesting that predictive analytics are particularly important in changing the balance of power between the plant and headquarters. The fact that results for predictive analytics are robust to inclusion of firm (as well as industry and county) fixed effects is important because it isolates within-firm variation in the use of predictive analytics across plants controlling for other changes

[29] For example, if the "hard" data received from predictive analytics complements "soft" information that is only available locally, then the hard data will increase the informational advantage of the local manager and, hence, delegation. If, on the other hand, predictive analytics substitutes for local expertise, it will tend to reduce the informational advantage of the local manager.

in underlying economics. In other words, within the firm, reductions in delegation are disproportionately concentrated in plants that have increased use of predictive analytics, making it more difficult to envision omitted variables that would explain the association between predictive analytics and delegation.

Based on the earlier discussion, these results suggest that predictive analytics primarily substitute for the private informational advantage of the plant manager, reducing the need for headquarters to delegate decision-making. For parsimony, we use a unidimensional factor in our primary analysis to capture the extent of delegation of discretionary authority over a range of decisions. However, the nature of the decision also likely affects the extent of delegation. In terms of the components of the delegation variable (not tabulated), the relation between changes in predictive analytics and delegation is driven by marketing (advertising, pricing and new product introduction) and human relations (hiring, promotion and pay), as opposed to capital expenditures. The result for marketing is consistent with the notion that predictive analytics permit headquarters to better forecast demand for the product and adjust advertising, pricing and product mix without relying on local expertise. Similarly, the result for human relations suggests that headquarters are better able to predict local staffing needs, and centralize hiring, promotion and pay. Consistent with these findings, later results on plant-level outcomes suggest that primary efficiency gains associated with predictive analytics relate to product mix, inventory and personnel utilization.

An alternate approach for understanding the relation between predictive analytics and delegation is to consider the source of demand for data from the plant. To the extent that the choice of specific data to be gathered is driven by managers at the plant, it seems more likely that the information is customized to their information needs, and therefore facilitates their decision-making. However, to the extent headquarters controls data gathering, it is more likely that the

resulting data facilitate centralized decision-making. In Table 4, Panel (b), we associate changes in predictive analytics with changes in responses to the question "who chose what type of data to gather at this establishment." In columns (1) and (3), we use an indicator if local management chose data types to be collected, whereas in columns (2) and (4), the dependent variable is an indicator if headquarters chose the type of data collected.[30] In columns (1) and (2), we see a significantly positive association between increases in *PAScore* and both local managers and headquarters choosing the type of data to collect at the establishment, consistent with a link between predictive analytics and increased data collection. More importantly, the increase in headquarters' involvement in determining the data collected (column (1)) is over three times as large as the increase in local management's involvement (column (2)), suggesting that data collection for predictive analytics is predominately initiated by headquarters. The fixed effects specifications in columns (3) and (4) yield similar conclusions with the increase only statistically significant for the headquarters' choice. Robustness to inclusion of firm fixed effects suggests that the increase in data gathering by headquarters is not firm-wide but, rather, is concentrated in plants using predictive analytics. Taken with the previous results, these findings suggest that predictive analytics usage is primarily controlled by headquarters and is used to centralize decision-making across a range of functions.

Predictive Analytics and Performance Evaluation Mechanisms

While better information can help centralize decision-making (the "decision-facilitating" role of information), it can also enable the principal to write contracts that better align incentives with the agent (the "decision-influencing" role). The preceding analyses suggest that insights from predictive analytics permit headquarters to centralize operational decisions (decision-facilitating).

[30] Responses are not mutually exclusive (respondents could select both local management and headquarters).

We next consider whether predictive analytics have a decision-influencing role for headquarters in constructing incentive contracts for plant managers. In particular, because predictive analytics increase information quality, they can provide objective metrics that correlate with future outcomes. We hence expect their implementation to be associated with greater use of explicit targets that focus on longer-term future outcomes deeper into the organization.

We begin by examining changes in the horizon of targets associated with predictive analytics. MOPS permits us to differentiate between short-term targets (less than a year) and longer-term targets. We expect the added forecasting ability from predictive analytics to be associated with a shift from short-term targets toward longer-term targets. Results in Table 5, Panel (a) are consistent with that prediction. Specifically, columns (1)-(4) provide strong evidence that increased use of predictive analytics is associated with a shift from short-term toward longer-term targets in compensating plant managers. In particular, predictive analytics usage is associated with a significant increase in the use of longer-term targets and a significant decrease in short-term targets. Results in columns (2) and (4), which include firm fixed effects, are particularly noteworthy because they indicate that the shift toward longer-term target use is not a firm-wide trend but, rather, is concentrated in the subset of plants that have adopted predictive analytics.

The MOPS also provides data on "who was aware of the production targets at this establishment." To the extent that targets become more important and permeate incentives lower into the organization, we expect awareness to increase. Results in columns (5)-(6) provide strong evidence that predictive analytics use is associated with target awareness deeper into the organization, consistent with targets receiving greater emphasis and, therefore, increasing is saliency. In particular, the relation between predictive analytics and target awareness is

significantly positive and robust to inclusion of firm fixed effects, indicating that the effect is specific to the subset of plants in a multi-plant firm that adopt predictive analytics.

Insert Table 5 here.

Next we turn to the "accuracy" of target setting. Prior research suggests that incentives lose their effectiveness if they are set either too low (because they are achieved with little effort) or too high (because they cannot be achieved even with great effort). If predictive analytics increase information quality, they can increase the accuracy of expectations, so we expect targets to be set more accurately. MOPS provides data on the question, "how easy or difficult was it for this establishment to achieve its production targets," with five categories of responses ranging from "possible to achieve without much effort" to "only possible to achieve with extraordinary effort." To the extent that predictive analytics permit headquarters to set more accurate targets as perceived by management ex ante, we expect fewer responses in the extremes and more toward the middle (particularly "normal" and "more than normal") effort. Furthermore, if these perceptions are borne out ex post, plants should more frequently meet or beat those expectations.

Results reported in Table 5, Panel (b), columns (1)-(5), are consistent with predictive analytics helping management set accurate targets. We observe significant negative coefficients for targets that are in the extremes (too easy or too hard), and significant positive coefficients on targets that require a "normal" or "more than normal" effort (i.e., realistic targets), suggesting that predictive analytics are associated with more accurate targets. However, results are insignificant if firm fixed effects are included (untabulated) suggesting that, while use of predictive analytics at the firm level increases target accuracy, we cannot differentiate the effect across specific plants. Column (6) reports results relating predictive analytics to the ability of managers to meet or beat

their targets. Consistent with the preceding results, expectations are, on average, accurate in that managers more frequently meet or beat targets with increased use of predictive analytics.[31]

Predictive Analytics and Incentive Intensity and Design

Finally, we turn to incentive intensity and design. To the extent that predictive analytics provide more accurate information on which to evaluate managers, we expect greater sensitivity of bonuses, promotion and firings to performance. In particular, we expect to see a positive relation between predictive analytics and use of bonuses, speed of terminations for underperformers, and promotions based on performance and ability rather than, "factors other than performance and ability (for example, tenure or family connections)."

Table 6, Panel (a), Columns (1) and (2) report results linking predictive analytics to use of performance-based bonuses. As expected, predictive analytics use is associated with a significant increase in use of performance-based bonuses. At the other extreme, to the extent that predictive analytics provide clear performance benchmarks, we expect firms to be more willing to fire underperforming plant managers because underperformance can be more accurately measured and dismissal can be more easily justified. Columns (3) and (4) report results linking predictive analytics to the propensity to fire managers for underperformance. Results suggest that use of predictive analytics increases the likelihood that managers will be fired or reassigned for underperformance, and that termination occurs more quickly. As with termination, we expect predictive analytics to be associated with a greater propensity to promote managers based on performance. Columns (5) and (6) provide evidence consistent with that prediction, suggesting that managers are more likely to receive performance-based promotions, as opposed to being

[31] This result may also reflect the possibility that, equipped with richer information, managers can adjust their effort to just meet the targets.

promoted based on "factors other than performance and ability (for example, tenure or family connections)." For all specifications in Table 6, Panel (a), results are similar using firm fixed effects, indicating that within-firm variation drives the relation between predictive analytics and incentives.

Insert Table 6 here.

The fact that predictive analytics are associated with stronger incentives is interesting given our earlier results documenting decreased delegation of decision-making. Prior theoretical research posits that incentive strength may increase when delegation *increases* because greater delegation implies a need to use stronger incentives to overcome the loss of control over the manager's actions (Prendergast, 2002). Our results suggest that predictive analytics provide a sufficient reduction in information asymmetry and increase in information quality to permit headquarters to centralize decisions (decision-facilitating) while maintaining strong incentives for plant managers (decision-influencing).

The preceding analyses indicate that predictive analytics are associated with reduced delegation and increased use of bonuses. However, it is unclear on what basis (plant or firm-wide) performance would be determined. In particular, while predictive analytics provide more plant-level data, our preceding results suggest that fewer decisions are delegated to the local manager. To the extent that decision-making has been centralized, we would expect a stronger link between bonuses and company-wide performance relative to plant-level measures. The MOPS provides data on whether manager bonuses were awarded based on their own performance or based on company-wide performance.

Table 6, Panel (b), presents results splitting bonuses between those awarded based on plant-level versus company-wide performance. Two points are worth noting. First, use of predictive

analytics is associated with increases in bonuses based on both company-wide (columns (1) and (2)) and plant-level performance (columns (3) and (4)). Second, and more interestingly, the point estimates for the increases in company-based bonuses in columns (1) and (2) are nearly twice as large as the increases in plant-based bonuses in columns (3) and (4).[32] Taken in conjunction with the prior results that headquarters chooses the data gathered, this analysis suggests that predictive analytics are used to centralize oversight of plant-level management, resulting in less discretion on the part of local managers and a commensurate increased link to company-wide performance.

Predictive Analytics and Government Mandates

Results to this point provide consistent evidence that increased use of predictive analytics at the plant level is associated with reductions in delegation and strengthening of incentive contracts tied to both plant and company-wide performance. From the firm's perspective, we view the increased sophistication and reduced cost of predictive analytics to be largely exogenous. The fact that our analysis is conducted in changes and conclusions are robust to the inclusion of firm, county, and industry fixed effects, along with time-varying controls, mitigates concerns about many potential omitted correlated variables. However, to draw stronger causal inference, we ideally would observe cases in which data collection was exogenously imposed on the plants.

One source of potential variation is changes in data collection resulting from government mandates. In particular, the MOPS asks respondents, "who chose what type of data to be collected at this establishment?" with a choice including "government regulations or agencies." To the extent that data collection was imposed by regulation or government agencies, it is largely outside the plant's control, but potentially has spillover effects that lessen the cost to collect and process

[32]Respondents check all that apply regarding the basis for management bonuses, so *CompanyBonus* and *PlantBonus* can both be one for a plant that has bonuses based both on company and plant performance criteria.

operational data. Thus, we should observe that government-mandated information collection is correlated with increases in the reliance on predictive analytics and performance indicators from production technology or instruments. Because government mandates could spill over to either of these information sources, we do not attempt to separate out these pathways, but instead sum Δ*PAScore* and Δ*ProdPerfIndScore* to create a new variable Δ*HardInfo*. We argue that, while government mandated information collection should affect Δ*HardInfo*, it is less likely to directly influence delegation, target setting, or incentive design since organizational architecture is under the control of the firm.[33]

To establish that government mandates do, in fact, affect the extent to which hard information is collected, we first estimate a regression in which Δ*HardInfo* is the dependent variable and changes in government mandates, plant payroll, and staff and management education levels are independent variables along with county and industry fixed effects. We estimate the model separately with and without firm fixed effects and separately for the smaller sample of remote plants used in our delegation analysis. Results for the first stage reported in Table 7, Panel (a) indicate that an increase in government mandates does, in fact, significantly increase the amount of hard data gathered by the firm.

In the second stage, we estimate our measures of delegation and targets as a function of the fitted variables from the first stage. Results, presented in Panels (b)-(d), are consistent with those reported earlier across all specifications. In particular, in terms of centralization, increased data gathering is associated with reductions in delegation to plant managers. In terms of incentive design, the increase in hard data is associated with increases in longer-term targets and awareness of targets throughout the organization. Further, increased hard data is associated with more

[33] Results are consistent if we substitute Δ*PAScore* for Δ*HardInfo*.

accurate targets as reflected in fewer that are set unreasonably high or low. Finally, in terms of incentive strength, increased hard data is associated with greater reliance on bonuses and stronger career outcome-based incentives. While we recognize that these results do not imply causality, it is comforting they are very similar to those in our primary analysis even when the source of demand for data is more likely to be exogenous to the firm.

Insert Table 7 here.

5. Plant-Level Outcomes and Predictive Analytics

In our final set of analyses, we explore the relation between predictive analytics and plant-level outcomes. As noted earlier, the Census data are limited in that they do not permit analysis of plant-level income statements. However, the ASM and CMF data contain some detailed information on plant operations, permitting us to observe if predictive analytics are associated with operational changes. To explore these relations, we build a panel of annual data for each plant from the ASM and CMF for 2008-2016, and conduct a difference-in-differences analysis, using increased reliance on predictive analytics as the treatment. We estimate models of the following structure:

$$Outcome_{pt} = \alpha_1\left(Post_t * \Delta PAScore_p\right) + \alpha_2\left(Post_t * \Delta ProdPerfIndScore_p\right) + \alpha_3\left(Post_t * \Delta MgrFdbckScore_p\right) + B \times Controls_{pt} + \gamma_p + \delta_{it} + \lambda_{ct} + \varepsilon_{pt}$$

where *p* indexes plant, *i* indexes the plant's 4-digit NAICS industry, *c* indexes the plant's county, and *t* indexes year. γ_p is a time-invariant plant fixed effect, δ_{it} is a time-varying industry fixed effect controlling for economic changes to the industry, and λ_{ct} is a time-varying county fixed effect controlling for local economic changes. $Post_t$ is an indicator that takes on a value of one after 2010.

Because the operational outcomes and some of the controls are captured on a yearly basis whereas predictive analytics are measured only in 2010 and 2015, we do not know precisely when the plant increased its use of predictive analytics. We adopt a lower-bound estimate and consider 2011-2016 to be in the post period.[34] α_1 is the difference-in-difference estimator for the effect of predictive analytics on plant outcomes. Because predictive analytics may be affected by firm-level decisions, we cluster standard errors at the firm level.

Because we do not observe overall profitability, we are limited to intermediate outcomes available in the Census data. In selecting outcome variables, we structure our analysis around the notion that, by providing continuous data and forecasts, predictive analytics facilitate a more flexible manufacturing system. In a flexible system, predictive data permit the plant to increase efficiency by anticipating demand and optimizing personnel and facilities usage (PMMI, 2018).

We expect several types of effects to be associated with predictive analytics. First, based on the delegation results in Table 4, we expect reductions in managerial payroll at the plant because the important decision-making and planning have been centralized to headquarters. Second, we expect changes to the lower-level staff complement. We have already documented significant increases in the education levels for the staff and managers in Table 3, consistent with the increased technological sophistication required to operate the plant. To facilitate greater optimization of facility usage, we expect increased reliance on workers who are cross-trained to work in multiple functions as demand changes. Consistent with greater flexibility, we expect an increased reliance on temporary workers and staff who operate on non-traditional schedules to keep production near capacity.

[34] If we limit the post period to 2015 and 2016 inferences are unchanged, but there are only two observations per plant in the post period.

Third, we expect the changes associated with predictive analytics to result in greater operating efficiency. In particular, we expect the plant to be able to generate greater output from a given level of capacity by utilizing predictive data and a flexible workforce to optimize production. To enhance efficiency, we expect a narrowing of the product mix to maximize production of the highest value-added products. In addition, we expect reductions in inventory levels (as a percentage of output) as production is synchronized more closely with demand.

Results in Table 8 provide evidence on workforce and operational changes associated with predictive analytics. Panels (a) and (b) report results for workforce changes. Results in Panel (a), Column (1) demonstrate that predictive analytics are associated with reductions in managerial payroll scaled by total value of shipments, consistent with greater centralization of decision-making. As documented in Column (2), staff payrolls scaled by shipments do not change significantly, suggesting that the primary effect is a reduction in the need for skilled plant managers.[35] Results in Column (3), indicate that predictive analytics are associated with greater reliance on temporary workers, consistent with a more flexible manufacturing process relying on a more fluid workforce. These specifications all control for changes in capital expenditures, the education of management and staff, and the number of distinct products shipped by the plant (coefficients suppressed for Census disclosure review purposes), providing comfort that our results are not driven by an unrelated structural change at the plant.

MOPS also provides data on the proportion of staff who are cross-trained and workers on flexible hours.[36] The results in Panel (b) provide detail on the use of flexible scheduling and cross-

[35] The fact that staff payrolls do not decrease is also reassuring because it suggests that the results for predictive analytics do not simply reflect greater automation through, for example, use of robotics.
[36] In Table 8, Panel (a), we use annual data from the ASM, but data for Panel (b) on flexible hours and cross-training are taken from MOPS and therefore are only available for 2010 and 2015. As a result, in Panel (b) we revert to a specification analogous to the previous tables.

trained employees, both with and without firm fixed effects. Results in Columns (1) and (2) indicate that flexible scheduling becomes more common with predictive analytics and Columns (3) and (4) indicate that a greater proportion of employees are cross-trained to work in multiple functions. Taken as a whole, the results suggest that predictive analytics are associated with a more flexible workforce that can respond nimbly to changes in demand as reflected in greater reliance on temporary workers, nontraditional scheduling and staff who can serve multiple roles.

Insert Table 8 here.

To the extent that plants re-optimize with predictive analytics, we expect results in terms of operational changes. Table, 8, Panel (c), reports operational changes associated with increases in predictive analytics. Columns (1) and (2) report changes in the total value of shipments and changes in the mix of products. Consistent with expectations, predictive analytics are associated with greater production volume from a given facility and with a streamlining in the range of products on offer, suggesting that enhanced analytics permit plants to optimize operations and streamline the product mix. Finally, results in Column (3) indicate enhanced efficiency as reflected in lower inventory levels as a percentage of volume shipped.[37]

Overall, results from Table 8 suggest that the availability of predictive analytics is associated with changes in plant-level outcomes in ways that are consistent with the results from our prior analyses. In particular, following adoption of predictive analytics, plants reduce their expenditures on local managers and increase their reliance on flexible staffing alternatives (temporary labor, flexible hours and cross-functional employees), resulting in increased use of productive capacity, streamlined product mix and more efficient inventory management. While

[37] In each of these specifications we control for changes in the staff and management education level at the plant and changes in capital expenditures. Columns (1) and (3) control for changes in the product offering at the plant, and (2) and (3) control for changes in staff and management wages.

we cannot infer that predictive analytics improve overall plant profitability, the results indicate significant predictable operating changes associated with their use.

6. Conclusion

Our analysis, while exploratory, suggests the following conclusions.

1. Predictive analytics have become increasingly pervasive in manufacturing, particularly for well-resourced firms with young plants and educated workforces in relatively stable industries.
2. In terms of organizational architecture, predictive analytics are associated with (a) reduced delegation of decision-making to plants, (b) increased reliance on, and more accurate, setting of performance targets, and (c) higher-powered incentives in tying compensation, promotion and firing more closely to performance outcomes.
3. In terms of operational changes, plants using predictive analytics increase overall production, streamline product mix and reduce inventory levels, while relying on a workforce that is increasingly temporary, flexible and cross-functional.

Our results are subject to important caveats. First, despite numerous controls, fixed effects, and alternative designs, we cannot draw strong causal conclusions. Second, we cannot draw normative conclusions because we are not able to observe the full range of outcomes or the costs of implementing a technologically-demanding information and control structure.

Subject to these caveats, predictive analytics and other big data techniques are an evolving force in manufacturing, and are likely to be increasingly important going forward as technology facilitates greater use of data for planning and optimization. While it is dangerous to extrapolate, our results suggest a future in which data analytics increasingly permit centralization of decision-

making to headquarters, with high-powered incentives for plant managers based on more precise performance measures and a push toward more temporary and irregular staff hiring similar to that observed in the service sector.

References

Agarwal, R., and Dhar, V. (2018). Editorial—Big Data, Data Science, and Analytics: The Opportunity and Challenge for IS Research. *Information Systems Research, 25*(3), 443-448. doi: 10.1287/isre.2014.0546

Aghion, P., and Tirole, J. (1997). Formal and Real Authority in Organizations. *Journal of Political Economy, 105*(1), 1-29.

Akturk, S. M., Ketzenberg, M., and Heim, G. R. (2018). Assessing impacts of introducing ship-to-store service on sales and returns in omnichannel retailing: A data analytics study. *Journal of Operations Management, 61*, 15-45. doi: 10.1016/j.jom.2018.06.004

Alles, M. G. (2015). Drivers of the Use and Facilitators and Obstacles of the Evolution of Big Data by the Audit Profession. *Accounting Horizons, 29*(2), 439-449. doi: 10.2308/acch-51067

Baesens, B., Bapna, R., Marsden, J. R., Vanthienen, J., and Zhao, J. L. (2016). Transformational Issues of Big Data and Analytics in Networked Business *MIS Quarterly, 40*(4), 807-818.

Bloom, N., Garicano, L., Sadun, R., and Van Reenen, J. (2014). The Distinct Effects of information Technology and Communication Technology on Firm Organization. *Management Science, 60*(12), 2859-2885.

Bouwens, J., Hofmann, C., and Van Lent, L. (2018). Performance Measures and Intra-Firm Spillovers: Theory and Evidence. *Journal of Management Accounting Research, forthcoming*.

Bouwens, J., and Kroos, P. (2017). The Interplay Between Forward-Looking Measures and Target Setting. *Management Science, 63*(9), 2868-2884. doi: 10.1287/mnsc.2016.2481

Bouwens, J., and Van Lent, L. (2007). Assessing the Performance of Business Unit Managers. *Journal of Accounting Research, 45*(4), 667-697.

Bresnahan, T. F., Brynjolfsson, E., and Hitt, L. M. (2002). Information Technology, Workplace Organization, and the Demand for Skilled Labor: Firm-Level Evidence. *Quarterly Journal of Economics*(Feb), 339-376.

Brickley, J. E., Smith, C. W., and Zimmerman, J. (1995). The Economics of Organizational Architecture. *Journal of Applied Corporate Finance, 8*(2), 19-31.

Brynjolfsson, E., Hitt, L. M., and Kim, H. H. (2011). Strength in Numbers: How Does Data-Driven Decisionmaking Affect Firm Performance? *Working Paper*, 33.

Brynjolfsson, E., and McElheran, K. (2016). The Rapid Adoption of Data-Driven Decision-Making. *American Economic Review, 106*(5), 133-139. doi: 10.1257/aer.p20161016

Bushman, R. M., Indjejikian, R. J., and Smith, A. (1995). Aggregate Performance Measures in Business Unit Manager Compensation: The Role of Intrafirm Interdependencies. *Journal of Accounting Research, 33*, 101-128.

Campbell, D. (2008). Nonfinancial Performance Measures and Promotion-Based Incentives. *Journal of Accounting Research, 46*(2), 297-332. doi: 10.1111/j.1475-679X.2008.00275.x

Campbell, D., Datar, S. M., and Sandino, T. (2009). Organizational Design and Control across Multiple Markets: the Case of Franchising in the Convenience Store Industry. *The Accounting Review, 84*(6), 1749-1779.

Corbett, C. J. (2018). How Sustainable Is Big Data? *Production and Operations Management, 27*(9), 1685-1695. doi: 10.1111/poms.12837

Cui, R., Gallino, S., Moreno, A., and Zhang, D. J. (2018). The Operational Value of Social Media Information. *Production and Operations Management, 27*(10), 1749-1769. doi: 10.1111/poms.12707

Deloitte. (2018). Exponential Technologies in Manufacturing: Transforming the Future of Manufacturing through Technology, Talent and the Innovation Ecosystem (pp. 64 pp).

Demski, J. S., and Feltham, G. A. (1976). *Cost Determination: A Conceptual Approach* (first ed.). Ames, Iowa: The Iowa State University Press.

Dilda, V., Mori, L., Noterdaeme, O., and Schmitz, C. (2017). Manufacturing: Analytics unleashes productivity and profitability (pp. 12): McKinsey & Company.

EY. (2014). Big Data Changing the way businesses compete and operate: EY.

Eyring, H., and Narayanan, V. G. (2018). Performance Effects of Setting a High Reference Point for Peer-Performance Comparison. *Journal of Accounting Research, 56*(2), 581-615. doi: 10.1111/1475-679x.12199

Furnham, A., and Boo, H. C. (2011). A literature review of the anchoring effect. *The Journal of Socio-Economics, 40*(1), 35-42. doi: 10.1016/j.socec.2010.10.008

Guha, S., and Kumar, S. (2018). Emergence of Big Data Research in Operations Management, Information Systems, and Healthcare: Past Contributions and Future Roadmap. *Production and Operations Management, 27*(9), 1724-1735. doi: 10.1111/poms.12833

Hecker, D. E. (2005). High-technology employment: a NAICS-based update. *Monthly Labor Review*(July), 57-72.

Henderson, R. M., and Clark, K. B. (1990). Architectural Innovation: The Reconfiguration of Existing Product Technologies and the Failure of Established Firms. *Administrative Science Quarterly, 35*(1), 9-30.

Hofmann, C., and Indjejikian, R. J. (2018, forthcoming). Authority and Accountability in Hierarchies - A Middle-Manager Perspective. *Foundations and Trends in Accounting*, 101.

Holmström, B. (1979). Moral Hazard and Observability. *Bell Journal of Economics, 10*(1), 74-91.

Ittner, C. D., and Larcker, D. F. (2001). Assessing Empirical Research in Managerial Accounting: a Value-Based Management Perspective. *Journal of Accounting and Economics, 32*, 349-410.

Moers, F. (2006). Performance Measure Properties and Delegation. *The Accounting Review, 81*(4), 897-924.

Nagar, V. (2002). Delegation and Incentive Compensation. *The Accounting Review, 77*(2), 379-395.

PMMI. (2018). *How to Utilize Big Data to Enhance Manufacturing Processes*. whitepaper.

Prendergast, C. (2002). The Tenuous Trade off between Risk and Incentives. *Journal of Political Economy, 110*(5), 1071-1102.

Ross, J. W., Beath, C. M., and Quaadgras, A. (2013). You May Not Need Big Data After All. *Harvard Business Review*, 90-98.

Sanders, N. R., and Ganeshan, R. (2018). Big Data in Supply Chain Management. *Production and Operations Management, 27*(10), 1745-1748. doi: 10.1111/poms.12892

Schneider, G. P., Dai, J., Janvrin, D. J., Ajayi, K., and Raschke, R. L. (2015). Infer, Predict, and Assure: Accounting Opportunities in Data Analytics. *Accounting Horizons, 29*(3), 719-742. doi: 10.2308/acch-51140

Singhal, K., Feng, Q., Ganeshan, R., Sanders, N. R., and Shanthikumar, J. G. (2018). Introduction to the Special Issue on Perspectives on Big Data. *Production and Operations Management, 27*(9), 1639-1641. doi: 10.1111/poms.12939

Tambe, P. (2014). Big Data Investment, Skills, and Firm Value. *Management Science, 60*(6), 1452-1469. doi: 10.1287/mnsc.2014.1899

Vasarhelyi, M. A., Kogan, A., and Tuttle, B. M. (2015). Big Data in Accounting: An Overview. *Accounting Horizons, 29*(2), 381-396. doi: 10.2308/acch-51071

Webb, R. A., Williamson, M. G., and Zhang, Y. (2013). Productivity-Target Difficulty, Target-Based Pay, and Outside-the-Box Thinking. *The Accounting Review, 88*(4), 1433-1457. doi: 10.2308/accr-50436

Appendix A: Variable Definitions

ChooseDataHQ	1 if respondent checked "Managers at headquarters and/or other establishments" for question 26 of the MOPS (who chose what type of data to collect at this establishment), 0 otherwise
ChooseDataLocal	1 if respondent checked "Managers at this establishment" for question 26 of the MOPS (who chose what type of data to collect at this establishment), 0 otherwise
CompanyBonus	1 if respondent checked "Their company's performance as measured by production targets" for managers' bonus criteria (question 11 of MOPS)
Crosstrain	Percent of employees who were cross-trained (question 43-4 of MOPS)
DelegationScore	Sum of questions 18-23 from MOPS (Various dimensions of delegation of decision-making). Responses to each question are scaled evenly so the response with the most autonomy is 1 and the least autonomy is 0
ExtremeEffortTarget	1 if respondent checked "Only possible to achieve with extraordinary effort" for question 7 of MOPS, 0 otherwise
FamilyCEO	1 if the respondent checked Yes to the CEO of the firm also being a founder or a member of the founder's family (question 45 of MOPS)
FirmAge	2015 minus year the firms oldest establishment first appeared in the Census Bureau's Longitudinal Business Database
FirmPayroll	Payroll of all establishments in the firm from the Census Bureau's Longitudinal Business Database
FlexibleHours	Percent of employees who were working flexible hours (question 43-2 of MOPS)
Government	1 if respondent checked "Government regulations or agencies" for question 26 of the MOPS (who chose what type of data to collect at this establishment), 0 otherwise
GrowthFirmPayroll	(2015 Firm Payroll - 2010 Firm Payroll)/2010 Firm Payroll
GrowthPlantPayroll	(2015 Plant Payroll - 2010 Plant Payroll)/2010 Plant Payroll
HardInfo	Sum of PAScore and ProdPerfIndScore
IndGrowth	5-year growth in TVS (on a percentage basis) for the 4-digit NAICS; obtained from aggregating ASM data using sampling weights
Inv	(Total Inventory at the Beginning of the year + Total Inventory at the End of the year)/(2 * TVS)
LittleEffortTarget	1 if respondent checked "Possible to achieve without much effort" for question 7 of MOPS, 0 otherwise
Margin	(TVS - Cost of Materials - Salaries and Wages)/TVS
MeetOrBeat	1 if respondent checked one of the first five responses to question 12 of MOPS, 0 otherwise

MgrFdbckScore	1 if respondent checked MOPS Question 27, second row as "Daily", 0.75 if "Weekly", 0.5 if "Monthly", 0.25 if "Yearly", 0 otherwise
MgtComp	Non-production workers salaries and wages scaled by TVS
MgtEducation	1 if respondent checked more than 80% of managers had a bachelors degree, 0.75 if 61-80%, 0.5 if 41-60%, 0.25 if 21-40% and 0 if 20% or less (question 40 of MOPS)
MoreThanNormalEffortTarget	1 if respondent checked "Possible to achieve with more than normal effort" for question 7 of MOPS, 0 otherwise
MultinationalFirm	1 if respondent checked the firm has production establishments in other countries (MOPS question 46), 0 otherwise
MultiplantFirm	1 if more than one establishment exists in the Census Bureau's Longitudinal Business Database for 2015
NormalEffortTarget	1 if respondent checked "Possible to achieve with normal effort" for question 7 of MOPS, 0 otherwise
NumProds	Number of distinct product class codes that the plant ships (https://www.census.gov/programs-surveys/asm/technical-documentation/product-codes-descriptions.html)
PAScore	1 if respondent checked MOPS Question 29 as "Daily", 0.75 if "Weekly", 0.5 if "Monthly", 0.25 if "Yearly", 0 otherwise
PlantAge	2015 minus year the establishment first appeared in the Census Bureau's Longitudinal Business Database
PlantBonus	1 if respondent checked "Their establishment's performance as measured by production targets" for managers' bonus criteria (question 11 of MOPS)
PlantPayroll	Payroll of the establishment from the Census Bureau's Longitudinal Business Database
Post	1 if year is 2015 or 2016, 0 otherwise
ProdPerfIndScore	1 if respondent checked MOPS Question 27, first row as "Daily", 0.75 if "Weekly", 0.5 if "Monthly", 0.25 if "Yearly", 0 otherwise
Promotion	Response to Question 14 of MOPS (the primary way managers were promoted at the establishment): 1="Promotions were based solely on performance and ability", 0.667="Promotions were based partly on performance and ability and partly on other factors (for example, tenure or family connections)", 0.333="Promotions were based mainly on factors other than performance and abillity", 0="Managers are normally not promoted"
RemotePlant	1 if respondent checked that headquarters for the company was not at the same location as this establishment (question 17 of MOPS), 0 otherwise
SomeEffortTarget	1 if respondent checked "Possible to achieve with some effort" for question 7 of MOPS, 0 otherwise
StaffComp	Production workers salaries and wages scaled by TVS

StaffEducation	1 if respondent checked more than 20% of non-managers had a bachelors degree, 0.667 if 11-20%, 0.333 if 1-10%, and 0 if 0% (question 41 of MOPS)
StaffWR	Production workers salaries and wages scaled by production workers' hours worked
TargetAwareness	Response to Question 8 of MOPS (who was aware of production targets at this establishment): 1="All managers and most production workers", 0.667="Most managers and most production workers", 0.333="Most managers and some production workers", 0="Only senior managers"
TCE	Total capital expenditures at the plant in a given year
TechInd	1 if the plant's primary NAICS is in one of the 4-Digit NAICS in table 1 of Hecker (2005)
TempEmp	Cost of temporary labor / Production workers salaries and wages
Termination	1 if respondent checked "within 6 months of identifying manager under-performance", 0.5 if "after 6 months of identifying manager under performance", or 0 if "Rarely or never" to question 16 of MOPS (speed of under-performing manager reassignment or dismissal)
TVS	Total value of shipments from the plant in a given year
UseBonus	1 if respondent checked any of the first 4 boxes to question 11 of MOPS (criteria for managerial bonuses)
UseLongTermTargets	1 if respondent checked "main focus was on long-term (more than one year) production targets" or "combination of short-term and long-term production targets"
UseMgrFdbck	1 if respondent checked any box for the second row of MOPS question 27 (Frequency of use of formal or informal feedback from managers) other than "Never", 0 otherwise
UsePA	1 if respondent checked any box for MOPS question 29 (Predictive analytics use frequency) other than "Never", 0 otherwise
UseProdPerfInd	1 if respondent checked any box for the first row of MOPS question 27 (Frequency of review of performance indicators from production technology or instruments) other than "Never", 0 otherwise
UseShortTermTargets	1 if respondent checked "main focus was on short-term (less than one year) production targets"

Table 1: Descriptive Statistics of ASM/CMF Data. Average values for plant-level variables from sample plants over 2008-2016. All variables are defined in Appendix A.

Variable	Mean
ln(TVS)	10.460
ln(NumProds)	1.180
MgtComp	0.071
StaffComp	0.105
StaffWR	21.170
Margin	0.307
Inv	0.117
TempEmp	0.065

Table 2: Descriptive Statistics of MOPS Data. Average values for 2015 levels and 2010 to 2015 changes in variables from MOPS. Levels are tabulated for the key constructs of interest and for the target accuracy variables. All variables are defined in Appendix A.

Variable	2015 Level	2010 to 2015 Change
UsePA	0.724	0.073
PAScore	0.478	0.087
ΔPAScore > 0	0.231	
UseProdPerfInd	0.896	0.034
ProdPerfIndScore	0.751	0.062
UseMgrFdbck	0.975	0.010
MgrFdbckScore	0.801	0.050
ChooseDataLocal		−0.003
ChooseDataHQ		0.039
DelegationScore		−0.111
UseLongTermTargets		0.172
UseShortTermTargets		−0.146
TargetAwareness		0.166
LittleEffortTarget	0.030	−0.010
SomeEffortTarget	0.105	−0.022
NormalEffortTarget	0.419	−0.007
MoreThanNormalEffortTarget	0.331	0.057
ExtremeEffortTarget	0.052	−0.016
MeetOrBeat	0.706	0.002
UseBonus		0.037
CompanyBonus		0.055
PlantBonus		0.016
Termination		0.082
Promotion		0.028
HardInfo		0.149
Government		0.014
Crosstrain		6.7
FlexibleHours		0.9

Table 3: Determinants of Predictive Analytics. This table presents regressions of predictive analytics (binary usage and frequency score variables) on firm and plant characteristics. Panel (a) tabulates 2015 levels analyses and panel (b) tabulates changes from 2010 to 2015. Panel (c) tabulates regressions of 2015 levels and 2010 to 2015 changes in predictive analytics on 2015 levels and 2010 to 2015 changes of other information sources (Performance indicators from production technology or instruments; formal or informal feedback from managers) and the respective controls used in panels (a) and (b) (control coefficients not tabulated). In panels (a) and (b), columns (3) and (4) include fixed effects for the plant's firm, 4-digit NAICS, and county, and in panel (c) fixed effects are included in columns (2), (4), (6), and (8). Standard errors are clustered at the firm level. All variables are defined in Appendix A.

(a) 2015 Levels

	UsePA	PAScore	UsePA	PAScore
	(1)	(2)	(3)	(4)
ln(FirmPayroll)	0.026***	0.024***		
	(0.002)	(0.003)		
MultinationalFirm	0.035***	0.028***		
	(0.008)	(0.008)		
FamilyCEO	−0.023***	−0.017***		
	(0.007)	(0.006)		
ln(FirmAge)	−0.031***	−0.021**		
	(0.010)	(0.009)		
MultiplantFirm	0.071***	0.042***		
	(0.009)	(0.008)		
IndGrowth	−0.049***	−0.080***		
	(0.015)	(0.013)		
TechInd	−0.049***	−0.066***		
	(0.008)	(0.007)		
RemotePlant	0.050***	0.054***	0.006	0.015
	(0.007)	(0.006)	(0.014)	(0.013)
ln(PlantAge)	−0.017***	−0.008	−0.007	−0.007
	(0.006)	(0.006)	(0.011)	(0.011)
ln(PlantPayroll)	0.031***	0.036***	0.022***	0.032***
	(0.003)	(0.003)	(0.006)	(0.006)
MgtEducation	0.019**	0.018**	−0.015	−0.001
	(0.009)	(0.008)	(0.019)	(0.017)
StaffEducation	0.097***	0.043***	0.056**	0.038***
	(0.011)	(0.009)	(0.023)	(0.021)
Constant	0.246***	−0.025		
	(0.033)	(0.026)		
N	26,500	26,500	26,500	26,500
R^2	0.13	0.149	0.783	0.718
Fixed Effects:				
Firm	No	No	Yes	Yes
Industry	No	No	Yes	Yes
County	No	No	Yes	Yes

(b) 2010 to 2015 Changes

	ΔUsePA (1)	ΔPAScore (2)	ΔUsePA (3)	ΔPAScore (4)
GrowthFirmPayroll	0.009**	0.012***		
	(0.004)	(0.003)		
ΔIndSalesGrowth	−0.003	−0.015***		
	(0.004)	(0.003)		
GrowthPlantPayroll	0.014***	0.004	0.01	0.003
	(0.004)	(0.003)	(0.009)	(0.007)
ΔMgtEducation	0.182***	0.200***	0.146***	0.163***
	(0.015)	(0.011)	(0.030)	(0.023)
ΔStaffEducation	0.164***	0.158***	0.154***	0.152***
	(0.017)	(0.012)	(0.037)	(0.028)
Constant	−0.321***	0.692***		
	(0.022)	(0.015)		
N	26,500	26,500	26,500	26,500
R^2	0.025	0.053	0.623	0.597
Fixed Effects:				
Firm	No	No	Yes	Yes
Industry	No	No	Yes	Yes
County	No	No	Yes	Yes

(c) Complementarities

	UsePA		PAScore		ΔUsePA		ΔPAScore	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
UseProdPerfInd	0.324***	0.263***						
	(0.012)	(0.038)						
UseMgrFdbck	0.138***	0.183***						
	(0.018)	(0.063)						
ProdPerfIndScore			0.253***	0.205***				
			(0.008)	(0.022)				
MgrFdbckScore			0.110***	0.128***				
			(0.009)	(0.024)				
ΔUseProdPerfInd					0.238***	0.195***		
					(0.014)	(0.035)		
ΔUseMgrFdbck					0.185***	0.221***		
					(0.023)	(0.060)		
ΔProdPerfIndScore							0.180***	0.170***
							(0.010)	(0.022)
ΔMgrFdbckScore							0.196***	0.201***
							(0.011)	(0.024)
N	26,500	26,500	26,500	26,500	26,500	26,500	26,500	26,500
R^2	0.184	0.790	0.214	0.731	0.077	0.638	0.156	0.638
Controls:	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fixed Effects:								
Firm	No	Yes	No	Yes	No	Yes	No	Yes
Industry	No	Yes	No	Yes	No	Yes	No	Yes
County	No	Yes	No	Yes	No	Yes	No	Yes

Table 4: Delegation of Decision Rights. Panel (a) tabulates regressions of changes in the amount of decision-making delegated to local plant management on changes in predictive analytics and other controls. The sample is all plants remote from corporate headquarters. Column (2) includes firm, industry, and county fixed effects. Panel (b) regresses changes in indicators of local management's (columns (1) and (3)) and headquarters management's (columns (2) and (4)) influence over the type of data collected on changes in predictive analytics intensity and other controls. The sample in this panel is the full complement of plants. Columns (3) and (4) include fixed effects for firm, industry, and county. Standard errors are clustered at the firm level. All variables are defined in Appendix A.

(a) Plant Management's Independence from Corporate Headquarters in Operational Decision-making

	ΔDelegationScore	
	(1)	(2)
ΔPAScore	−0.183***	−0.136**
	(0.043)	(0.064)
ΔProdPerfIndScore	−0.139***	−0.088
	(0.051)	(0.067)
ΔMgrFdbckScore	0.070	0.105
	(0.056)	(0.083)
Growth FirmPayroll	−0.091***	
	(0.017)	
ΔIndSalesGrowth	0.025	
	(0.016)	
Growth PlantPayroll	0.083***	0.070***
	(0.016)	(0.026)
ΔMgtEducation	−0.084	−0.031
	(0.053)	(0.073)
ΔStaffEducation	−0.093	−0.155*
	(0.057)	(0.085)
Constant	6.369***	
	(0.103)	
N	11,500	11,500
R^2	0.014	0.524
Fixed Effects:		
Firm	No	Yes
Industry	No	Yes
County	No	Yes

(b) Choice of Information Collection

	ΔChooseDataLocal (1)	ΔChooseDataHQ (2)	ΔChooseDataLocal (3)	ΔChooseDataHQ (4)
ΔPAScore	0.031***	0.100***	0.039	0.108***
	(0.011)	(0.013)	(0.024)	(0.030)
ΔProdPerfIndScore	0.001	0.071***	−0.024	0.093***
	(0.012)	(0.013)	(0.031)	(0.034)
ΔMgrFdbckScore	0.051***	0.061***	0.041	0.069*
	(0.015)	(0.016)	(0.037)	(0.040)
GrowthFirmPayroll	−0.010**	0.023***		
	(0.004)	(0.005)		
ΔIndSalesGrowth	0.001	−0.009**		
	(0.004)	(0.004)		
GrowthPlantPayroll	0.009**	−0.013***	0.005	−0.005
	(0.004)	(0.005)	(0.009)	(0.010)
ΔMgtEducation	0.009	0.061***	0.01	0.069**
	(0.012)	(0.013)	(0.028)	(0.030)
ΔStaffEducation	−0.005	0.043***	−0.018	0.062
	(0.014)	(0.015)	(0.035)	(0.039)
Constant	−0.093***	−0.332***		
	(0.023)	(0.025)		
N	26,500	26,500	26,500	26,500
R^2	0.004	0.028	0.470	0.482
Fixed Effects:				
Firm	No	No	Yes	Yes
Industry	No	No	Yes	Yes
County	No	No	Yes	Yes

Table 5: Targets. Panel (a) tabulates regressions of changes in characteristics of the targets used at the plant on changes in predictive analytics and other controls. The dependent variable in columns (1) and (2) is changes in an indicator of the use of targets that include a long-term component, in columns (3) and (4) is changes in an indicator of the use of only short-term targets, and in (5) and (6) is changes in a score of how much lower level employees are aware of the plant's targets. Columns (2), (4) and (6) include firm fixed effects for firm, industry, and county. Panel (b) tabulates regressions of changes in indicators of target difficulty and the likelihood of achieving targets on changes in predictive analytics and other controls. Fixed effects for the plant's industry and county are used in panel (b). In both panels standard errors are clustered at the firm level. All variables are defined in Appendix A.

(a) Target Characteristics

	ΔUseLongTermTargets		ΔUseShortTermTargets		ΔTargetAwareness	
	(1)	(2)	(3)	(4)	(5)	(6)
ΔPAScore	0.455***	0.377***	−0.367***	−0.300***	0.424***	0.368***
	(0.018)	(0.040)	(0.019)	(0.043)	(0.013)	(0.028)
ΔProdPerfIndScore	0.215***	0.206***	−0.108***	−0.112**	0.203***	0.201***
	(0.018)	(0.042)	(0.020)	(0.044)	(0.013)	(0.031)
ΔMgrFdbckScore	0.226***	0.236***	−0.133***	−0.179***	0.214***	0.195***
	(0.021)	(0.050)	(0.024)	(0.054)	(0.015)	(0.036)
Growth FirmPayroll	0.016**		−0.017**		0.018***	
	(0.007)		(0.007)		(0.004)	
ΔIndSalesGrowth	−0.007		0.01		−0.016***	
	(0.006)		(0.007)		(0.005)	
Growth PlantPayroll	0.012**	0.021	−0.002	−0.012	0.003	0.013
	(0.006)	(0.014)	(0.007)	(0.015)	(0.004)	(0.009)
ΔMgtEducation	0.182***	0.217***	−0.149***	−0.179***	0.180***	0.169***
	(0.019)	(0.041)	(0.020)	(0.043)	(0.014)	(0.031)
ΔStaffEducation	0.161***	0.118**	−0.140***	−0.097*	0.121***	0.094***
	(0.022)	(0.052)	(0.024)	(0.057)	(0.015)	(0.036)
Constant	−1.186***		0.836***		−0.077***	
	(0.030)		(0.036)		(0.021)	
N	26,500	26,500	26,500	26,500	26,500	26,500
R^2	0.142	0.615	0.071	0.592	0.236	0.671
Fixed Effects:						
Firm	No	Yes	No	Yes	No	Yes
Industry	No	Yes	No	Yes	No	Yes
County	No	Yes	No	Yes	No	Yes

(b) Target Difficulty - Changes

	ΔEffort Required to Achieve Target					ΔMeet or
	Little	Some	Normal	More than Normal	Extreme	Beat
	(1)	(2)	(3)	(4)	(5)	(6)
ΔPAScore	−0.035***	−0.031*	0.107***	0.085***	−0.121***	0.018**
	(0.010)	(0.017)	(0.023)	(0.023)	(0.014)	(0.009)
ΔProdPerfIndScore	−0.008	0.003	0.081***	−0.01	−0.063***	0.026***
	(0.010)	(0.016)	(0.022)	(0.023)	(0.014)	(0.009)
ΔMgrFdbckScore	0.005	0.041**	0.026	0.04	−0.110***	0.046***
	(0.012)	(0.018)	(0.026)	(0.026)	(0.017)	(0.011)
Growth FirmPayroll	−0.002	−0.005	0.018**	−0.007	−0.003	−0.00001
	(0.003)	(0.005)	(0.008)	(0.008)	(0.005)	(0.003)
Growth PlantPayroll	−0.003	0.002	0.008	−0.004	−0.003	0.005
	(0.003)	(0.005)	(0.008)	(0.008)	(0.005)	(0.004)
ΔMgtEducation	−0.019*	0.027	0.014	0.002	−0.021	0.022**
	(0.010)	(0.017)	(0.024)	(0.024)	(0.016)	(0.011)
ΔStaffEducation	−0.009	0.007	−0.009	0.021	−0.013	0.006
	(0.011)	(0.020)	(0.029)	(0.030)	(0.017)	(0.013)
N	26,500	26,500	26,500	26,500	26,500	26,500
R^2	0.021	0.019	0.022	0.019	0.042	0.030
Fixed Effects:						
Firm	No	No	No	No	No	No
Industry	Yes	Yes	Yes	Yes	Yes	Yes
County	Yes	Yes	Yes	Yes	Yes	Yes

Table 6: Managerial Incentives. Panel (a) tabulates regressions of changes in the use of bonuses, termination-based incentives, and promotion-based incentives used at the plant on changes in predictive analytics and other controls. Columns (2), (4) and (6) include firm fixed effects for firm, industry, and county. Panel (b) tabulates regressions of changes in indicators of company performance based bonuses and plant performance based bonuses on changes in predictive analytics and other controls. In this panel, Columns (2) and (4) include firm fixed effects for firm, industry, and county. Fixed effects for the plant's industry and county are used in panel (b). In both panels standard errors are clustered at the firm level. All variables are defined in Appendix A.

(a) Incentive Strength

	ΔUseBonus		ΔTermination		ΔPromotion	
	(1)	(2)	(3)	(4)	(5)	(6)
ΔPAScore	0.091***	0.070***	0.223***	0.202***	0.084***	0.073***
	(0.012)	(0.026)	(0.013)	(0.029)	(0.008)	(0.016)
ΔProdPerfIndScore	0.08***	0.065**	0.111***	0.105***	0.063***	0.076***
	(0.013)	(0.028)	(0.013)	(0.032)	(0.009)	(0.018)
ΔMgrFdbckScore	0.082***	0.042	0.210***	0.196***	0.090***	0.073***
	(0.015)	(0.032)	(0.017)	(0.038)	(0.011)	(0.023)
Growth FirmPayroll	0.016***		0.007*		0.006**	
	(0.004)		(0.004)		(0.003)	
ΔIndSalesGrowth	0.021***		−0.013***		0.001	
	(0.005)		(0.004)		(0.002)	
Growth PlantPayroll	0.017***	0.009	−0.003	0.005	−0.002	0.0003
	(0.004)	(0.009)	(0.004)	(0.009)	(0.002)	(0.005)
ΔMgtEducation	0.016	0.035	0.149***	0.163***	0.071***	0.081***
	(0.013)	(0.028)	(0.014)	(0.033)	(0.009)	(0.020)
ΔStaffEducation	0.037**	0.025	0.077***	0.072*	0.057***	0.037
	(0.015)	(0.033)	(0.016)	(0.037)	(0.011)	(0.024)
Constant	−0.341***		0.259***		0.634***	
	(0.024)		(0.024)		(0.018)	
N	26,500	26,500	26,500	26,500	26,500	26,500
R^2	0.028	0.671	0.110	0.600	0.072	0.642
Fixed Effects:						
Firm	No	Yes	No	Yes	No	Yes
Industry	No	Yes	No	Yes	No	Yes
County	No	Yes	No	Yes	No	Yes

(b) Incentive Criteria

	ΔCompanyBonus		ΔPlantBonus	
	(1)	(2)	(3)	(4)
ΔPAScore	0.127***	0.106***	0.071***	0.060*
	(0.015)	(0.034)	(0.013)	(0.030)
ΔProdPerfIndScore	0.089***	0.092***	0.032***	0.031
	(0.015)	(0.036)	(0.012)	(0.031)
ΔMgrFdbckScore	0.070***	0.034	0.041***	0.063*
	(0.018)	(0.040)	(0.015)	(0.036)
Growth FirmPayroll	0.023***		−0.003	
	(0.006)		(0.005)	
ΔIndSalesGrowth	0.008		0.001	
	(0.005)		(0.005)	
Growth PlantPayroll	−0.001	0.008	0.012**	0.007
	(0.005)	(0.011)	(0.005)	(0.010)
ΔMgtEducation	0.060***	0.057	0.041***	0.047
	(0.016)	(0.036)	(0.013)	(0.032)
ΔStaffEducation	0.036*	0.04	0.026*	0.042
	(0.019)	(0.044)	(0.016)	(0.041)
Constant	−0.385***		−0.220***	
	(0.029)		(0.024)	
N	26,500	26,500	26,500	26,500
R^2	0.025	0.563	0.010	0.448
Fixed Effects:				
Firm	No	Yes	No	Yes
Industry	No	Yes	No	Yes
County	No	Yes	No	Yes

Table 7: Government-Mandated Data Collection

(a) First Stage: Government Mandates driving Hard Information Use. Panel (a) tabulates regressions of the change in hard information collected on changes in an indicator for government regulations or agencies contributing to the selection of the type of data collected and other control variables. Column (1) uses plant industry and county fixed effects whereas columns (2) and (3) add firm fixed effects. Columns (1) and (2) are estimated on the full sample, and column (3) is estimated on the sample of remote plants used in Table 4(a). Standard errors are clustered at the firm level. All variables are defined in Appendix A.

	ΔHardInfo		
	(1)	(2)	(3)
ΔGovernment	0.164***	0.166***	0.130***
	(0.020)	(0.037)	(0.037)
ΔMgrFdbckScore	0.904***	0.890***	0.842***
	(0.019)	(0.039)	(0.040)
Δln(PlantPayroll)	0.015***	0.0001	0.002
	(0.004)	(0.011)	(0.011)
ΔMgtEducation	0.236***	0.185***	0.199***
	(0.016)	(0.034)	(0.034)
ΔStaffEducation	0.186***	0.171***	0.155***
	(0.017)	(0.038)	(0.039)
N	26,500	26,500	11,500
R^2	0.313	0.714	0.605
Fixed Effects:			
Firm	No	Yes	Yes
Industry	Yes	Yes	Yes
County	Yes	Yes	Yes
Sample:	Main		Delegation

(b) Second Stage: Delegation and Targets. This panel tabulates regressions of changes in delegation and target characteristics on fitted values of ΔHardInfo and the control (non-government) variables from panel (a). Column (1) uses fitted values from the model in column (3) of panel (a); columns (2)-(4) use fitted values from the model in column (2) of panel (a). Each column includes firm, plant industry, and county fixed effects. Standard errors are clustered at the firm level. All variables are defined in Appendix A.

	ΔDelegation	ΔUseLongTermTargets	ΔUseShortTermTargets	ΔTargetAwareness
	(1)	(2)	(3)	(4)
$\Delta\widehat{\text{HardInfo}}$	-1.756^{**}	0.815^{***}	-0.623^{*}	0.815^{***}
	(0.869)	(0.310)	(0.337)	(0.215)
N	11,500	26,500	26,500	26,500
R^2	0.524	0.598	0.583	0.641
Controls:	Yes	Yes	Yes	Yes
Fixed Effects:				
Firm	Yes	Yes	Yes	Yes
Industry	Yes	Yes	Yes	Yes
County	Yes	Yes	Yes	Yes

(c) Second Stage: Target Difficulty. This panel tabulates regressions of changes in target difficulty and meet-or-beat indicators on fitted values of ΔHardInfo and the control (non-government) variables from panel (a). Each column uses fitted values from the model in column (1) of panel (a) and employs the plant's industry and county fixed effects. Standard errors are clustered at the firm level. All variables are defined in Appendix A.

	ΔEffort Required to Achieve					ΔMeet or
	Little	Some	Normal	More than Normal	Extreme	Beat
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta\widehat{\text{HardInfo}}$	-0.187^{**}	0.032	0.375^{**}	-0.006	-0.198^{*}	0.054
	(0.093)	(0.128)	(0.185)	(0.194)	(0.116)	(0.073)
N	26,500	26,500	26,500	26,500	26,500	26,500
R^2	0.02	0.019	0.02	0.018	0.034	0.029
Controls:	Yes	Yes	Yes	Yes	Yes	Yes
Fixed Effects:						
Firm	No	No	No	No	No	No
Industry	Yes	Yes	Yes	Yes	Yes	Yes
County	Yes	Yes	Yes	Yes	Yes	Yes

(d) Second Stage: Incentives. This panel tabulates regressions of changes in bonus and career-path incentives on fitted values of ΔHardInfo and the control (non-government) variables from panel (a). Columns (1), (3), (5) and (7) use fitted values from the model in column (1) of panel (a) and employ the plant's industry and county fixed effects. Columns (2), (4), (6) and (8) use fitted values from the model in column (2) of panel (a) and include firm, plant industry, and county fixed effects. Standard errors are clustered at the firm level. All variables are defined in Appendix A.

	ΔPlantBonus		ΔCompanyBonus		ΔTerminations		ΔPromotions	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta\widehat{\text{HardInfo}}$	0.435***	0.317	0.305***	0.215	0.440***	0.217	0.259***	0.298*
	(0.127)	(0.278)	(0.116)	(0.264)	(0.107)	(0.217)	(0.080)	(0.152)
N	26,500	26,500	26,500	26,500	26,500	26,500	26,500	26,500
R^2	0.039	0.560	0.027	0.447	0.113	0.590	0.084	0.636
Controls:	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fixed Effects:								
Firm	No	Yes	No	Yes	No	Yes	No	Yes
Industry	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
County	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Table 8: Outcomes

(a) Workforce Changes: ASM/CMF Data. The sample for this panel is the MOPS data from the main sample linked with ASM/CMF data for each plant from 2008-2016. The dependent variable in column (1) is plant non-production worker salaries and wages scaled by total value of shipments, column (2) is plant production worker salaries and wages scaled by total value of shipments, and column (3) is plant expenditures on temporary employees scaled by production worker salaries and wages. $Post_t$ is an indicator if the year for the ASM/CMF data is 2011-2016. Each column includes control variables (coefficients untabulated) for $\text{Post}_t \times \Delta\text{MgtEducation}_p$, $\text{Post}_t \times \Delta\text{StaffEducation}_p$, $\ln(\text{NumProds}_p)$, and $\text{TCE}_p/\text{TVS}_p$. These specifications include plant fixed effects, plant industry × year fixed effects, and plant county × year fixed effects. Standard errors are clustered at the firm level. All variables are defined in Appendix A.

	MgtComp (1)	StaffComp (2)	TempEmp (3)
Post × ΔPAScore	−0.003**	−0.0001	0.008***
	(0.001)	(0.001)	(0.003)
Post × ΔProdPerfIndScore	−0.002	−0.001	0.001
	(0.001)	(0.001)	(0.003)
Post × ΔMgrFdbckScore	0.001	0.001	0.0004
	(0.001)	(0.002)	(0.003)
N	180,000	180,000	180,000
R^2	0.857	0.880	0.703
Fixed Effects:			
Plant	Yes	Yes	Yes
Industry × Year	Yes	Yes	Yes
County × Year	Yes	Yes	Yes
Other Controls:	Yes	Yes	Yes

(b) Workforce Changes: MOPS Data. The sample for this panel is the MOPS data sample used in the main analyses. The dependent variable in columns (1) and (2) is the difference in the percentage of employees from 2010 to 2015 that work flexible hours. In columns (3) and (4), the dependent variable is the difference in the percentage of employees that are crosstrained. Columns (2) and (4) include firm, plant industry, and plant county fixed effects. Standard errors are clustered at the firm level. All variables are defined in Appendix A.

	ΔFlexibleHours		ΔCrosstrained	
	(1)	(2)	(3)	(4)
ΔPAScore	1.282***	1.107**	10.05***	9.086***
	(0.263)	(0.554)	(0.589)	(1.340)
ΔProdPerfIndScore	0.341	−0.078	5.170***	5.149***
	(0.235)	(0.548)	(0.595)	(1.366)
ΔMgrFdbckScore	−0.082	0.331	6.468***	6.620***
	(0.377)	(0.644)	(0.726)	(1.675)
Growth FirmPay	−0.167*		0.278	
	(0.087)		(0.201)	
ΔIndSalesGrowth	−0.434***		−0.217	
	(0.108)		(0.206)	
Growth PlantPay	−0.016	0.116	−0.498**	−0.497
	(0.086)	(0.225)	(0.218)	(0.506)
ΔMgtEducation	1.084***	0.689	9.245***	8.670***
	(0.384)	(0.862)	(0.748)	(1.664)
ΔStaffEducation	1.451***	1.281	9.928***	10.34***
	(0.426)	(1.055)	(0.847)	(2.038)
Constant	−3.09***		−3625***	
	(0.666)		(1.226)	
N	26,500	26,500	26,500	26,500
R^2	0.006	0.560	0.112	0.597
Fixed Effects:				
Firm	No	Yes	No	Yes
Industry	No	Yes	No	Yes
County	No	Yes	No	Yes

(c) Operational Changes. The sample for this panel is the MOPS data from the main sample linked with ASM/CMF data for each plant from 2008-2016. The dependent variable in column (1) is the natural logarithm of total value of shipments in a given year, column (2) is the natural logarithm of number of distinct product class codes the plant ships in a given year, and column (3) is average inventories at the plant scaled by total value of shipments. $Post_t$ is an indicator if the year for the ASM/CMF data is 2011-2016. Each column includes control variables (coefficients untabulated) for $\text{Post}_t \times \Delta\text{MgtEducation}_p$ and $\text{Post}_t \times \Delta\text{StaffEducation}_p$. Columns (1) and (3) include $\ln(\text{NumProds}_p)$ as a control. Column (2) includes a control for $\ln(\text{TVS}_p)$. Column (1) includes a control for $\ln(1 + \text{TCE}_p)$ whereas columns (2) and (3) include the scaled version $\text{TCE}_p/\text{TVS}_p$. Columns (2) and (3) include MgtComp_{pt} and StaffComp_{pt} as controls. All specifications include plant fixed effects, plant industry × year fixed effects, and plant county × year fixed effects. Standard errors are clustered at the firm level. All variables are defined in Appendix A.

	ln(TVS)	ln(NumProds)	Inv
	(1)	(2)	(3)
Post × ΔPAScore	0.051***	−0.045**	−0.004**
	(0.019)	(0.019)	(0.002)
Post × ΔProdPerfIndScore	0.032	−0.010	0.005**
	(0.021)	(0.020)	(0.002)
Post × ΔMgrFdbckScore	−0.024	0.0004	−0.003
	(0.025)	(0.024)	(0.002)
N	180,000	180,000	180,000
R^2	0.964	0.643	0.864
Plant Fixed Effects:	Yes	Yes	Yes
Industry × Year Fixed Effects:	Yes	Yes	Yes
County × Year Fixed Effects:	Yes	Yes	Yes
Other Controls Included:	Yes	Yes	Yes